KeyVault / Huskey Pocket

Technical Feature Specification and Operational Catalog

This document summarizes the features of Huskey Pocket, a credential manager based on the Waveshare ESP32-S3-Touch-LCD-2.8 board. The device works without cloud services and can send credentials over USB or Bluetooth when requested by the user.

1. Overview and Hardware Architecture

- **Microcontroller: Espressif ESP32-S3 dual-core Xtensa LX7 at 240 MHz, with integrated PSRAM and Flash memory.**
- **Display: 2.8-inch IPS TFT LCD with 240 × 320 resolution and an ST7789 controller connected over a high-speed SPI bus.**
- **Touch digitizer: CST328 capacitive sensor on I2C (Wire1), with gesture support and continuous updates.**
- **Real-time clock (RTC): the PCF85063 requires an external backup battery to retain the time while the device is off.**
- **Inertial sensor (IMU): six-axis QMI8658C accelerometer and gyroscope module over I2C (Wire) for measuring motion and orientation vectors.**
- **Removable storage: microSD card slot using the 1-bit SD_MMC protocol for file transfers.**
- **Haptic feedback: ERM vibration motor driven by a transistor to provide tactile feedback for key presses and interface events.**
- **Power management: LiPo cell voltage measurement through a calibrated ADC, peripheral power cutoff through a P-channel MOSFET (AO3401), and deep sleep.**

2. Security and Cryptographic Vault

- **Vault encryption: entry fields are encrypted with AES-256-CBC through mbedTLS. CBC does not provide built-in authentication.**
- **Key derivation: PBKDF2-HMAC-SHA256 with 50,000 iterations for the vault PIN. .hky backups use 100,000 iterations and an export password.**
- **PIN: six-digit numeric code.**
- **Brute-force protection: progressive temporary lockouts after consecutive failed attempts.**
- **Secure storage: encrypted records reside in internal Flash using LittleFS, with operational settings stored in NVS (Non-Volatile Storage).**
- **Factory reset: removes vault data and local preferences; the FRP PIN remains to prevent reconfiguration without verification. Logical deletion does not guarantee physical overwriting of every Flash cell.**

3. Active Protection Features

- **Snatch Lock: while the vault is open, the firmware polls the QMI8658C IMU and may lock the device when acceleration and rotation exceed the selected thresholds. Detection is software-based; no fixed response time is guaranteed.**
- **Lost Mode: lets the user select a period in days; if it expires without a genuine unlock, the firmware starts a factory reset. It depends on the RTC retaining the time.**

- **Auto-lock on inactivity: an inactivity timer turns off the display and requires the PIN again after the selected period.**

4. HID Keyboard Emulation (Credential Typing)

- **USB HID mode: identifies itself to the host as a standard keyboard through the native TinyUSB stack.**
- **Bluetooth Low Energy (BLE) mode: wireless keyboard emulation using NimBLE-Arduino, compatible with Windows, macOS, Linux, iOS, and Android, with encrypted bonding support.**
- **Keyboard layouts: supports ES-LAT (Latin American Spanish), ES-ES (Spain Spanish), and US (United States English), including accents, special characters, and capitalization.**
- **One-touch typing: shortcuts send Username + Tab, Password + Enter, notes text, and generated TOTP tokens.**

5. Entry and Credential Management

- **Entry types: Web Accounts (Title, Username, Password, Notes/URL, TOTP) and Secrets / Notes Only (multiline free text for seed phrases or private keys).**
- **Category structure: organize entries into folders and categories; create, rename, and delete folders without losing their entries.**
- **Favorites: mark entries for quick access from the main screen.**
- **Live search: progressively filters results as text is entered on the keyboard.**
- **Protected display: passwords are masked, with on-demand reveal and haptic confirmation.**
- **QR code sharing: displays high-contrast QR codes that can be scanned directly by phones without a network connection.**

6. Two-Factor Authentication (TOTP / 2FA)

- **Standard: compatible with RFC 6238 (HMAC-SHA1, six digits, 30-second time step), including Google Authenticator and similar apps.**
- **Time accuracy: continuous timekeeping relies on the hardware clock and the external PCF85063 RTC. The RTC requires an external backup battery to retain time while the device is off.**
- **Dynamic display: graphical and numeric countdown refreshed every second.**
- **Typing: a button sends the 2FA code to the host as keyboard input.**

7. Built-in Password Generator

- **Generator: password length from 8 to 32 characters.**
- **Character sets: independent options for uppercase, lowercase, numbers, and special symbols.**
- **Ambiguous-character filter: optional exclusion of visually confusable characters (1, l, I, 0, O).**
- **Access modes: available as a quick password-generation screen or embedded in create and edit forms.**

8. On-screen Virtual Keyboards

- **Split keyboard: six sliding panels designed for the 2.8-inch screen, with one-shot Shift, Caps Lock, and panel navigation.**
- **T9 keyboard (4×4 grid): 16-key layout with multitap and long-press input for direct numbers and shortcuts such as .com. Includes an OK key to confirm text.**

9. Storage, Backups, and Data Transfer

- **Encrypted backups (.hky): exports the complete vault to microSD in a password-protected backup using PBKDF2 and AES-256.**
- **CSV import and export: uses the microSD card; CSV files remain readable as plain text.**
- **Import: processes .hky and .csv files in steps to keep the interface responsive. microSD initialization also runs in the background after the access screen appears.**
- **Managed files in /tmp and /export can be cleaned up. Overwriting data in Flash does not guarantee physical removal of every copy.**
- **USB Mass Storage (MSC) mode: exposes the microSD card directly to a computer as removable storage.**

10. User Interface and Experience (LVGL 8.x)

- **Status bar: segmented battery indicator in 25% increments with a charging animation, real-time digital clock, and SD-card and lock indicators.**
- **Themes: Dark, Light, and Custom themes with hue and saturation controls. The interface calculates luminance to select contrasting text colors.**
- **Haptic feedback: configurable vibration for touch-key presses, unlocks, and system confirmations.**
- **Toast notifications: brief confirmations of actions that do not interrupt navigation.**
- **Power menu: floating circular panel for shutdown, restart, or manual lock.**
- **Onboarding: first-start setup flow for setting the master PIN, language, date, and time.**
- **Multilingual support: translated strings are integrated for offline operation.**